

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

MLOPs Report

Group ID: 25-26J-091

Project Title: Smart Real-Time Age, Gender, Emotion, and Behavior-Based Advertising System for Sri Lankan Malls

1. IT22109712: Age and Gender Detection
2. IT22258526: Emotion Detection
3. IT22184030: Behavioral Engagement Detection
4. IT22071934: Advertisement Recommendation

		IT22109712	IT22258526	IT22184030	IT22071934
Data Pipeline	Data Sources:	The age and gender detection model uses publicly available datasets such as UTKFace and IMDB-WIKI. These datasets contain facial images labeled with age and gender, which are suitable for supervised learning.	Emotion detection uses Sri lankan facial emotion dataset which contain facial expressions labeled with emotions like happy, sad, angry, fear and neutral.	This model uses simulated or derived datasets based on video streams. Features such as gaze direction, head orientation, and interaction are extracted using real-time data.	A custom dataset stored in MongoDB is used, containing advertisement metadata such as target age group, gender, mood, and category.
	Data Preprocessing:	Images are resized	Images are resized,	Video frames are	Data is cleaned,

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		to a fixed resolution (e.g., 224x224) and normalized. Face detection is performed using MTCNN to extract relevant facial regions. Data augmentation techniques such as flipping, rotation, and brightness adjustment are applied to improve model generalization.	normalized, and augmented. Facial regions are extracted using MTCNN. Class imbalance is handled using weighted loss functions or oversampling techniques.	processed using OpenCV. YOLO is used for object detection, and tracking is done using SORT. Features such as attention percentage are calculated from detected behavior	missing values are handled, and categorical variables are standardized. Data is converted into structured formats such as DataFrames.
	Data storage and versioning	Training datasets are stored locally, while processed data can be managed using tools such as DVC for versioning. Git is used for code version control to	Data is stored locally, and versioning is maintained using Git and optionally DVC for dataset tracking	Processed engagement data and logs are stored in MongoDB. Versioning is handled through Git	Advertisement data is stored in MongoDB. Versioning can be maintained using Git and database backups.

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		ensure reproducibility.			
Model Development	Model Selection	A Convolutional Neural Network (CNN) built using TensorFlow/Keras is used due to its effectiveness in image classification tasks.	A CNN-based classification model is used to detect multiple emotional states from facial expressions.	A combination of YOLO for object detection and a custom CNN for head orientation is used.	A rule-based filtering system combined with similarity-based methods (e.g., embeddings using Sentence Transformers) is used for recommendation.
	Model Training	The dataset is split into training, validation, and testing sets. The model is trained using categorical cross-entropy loss and evaluated using accuracy and F1-score. Hyperparameters such as learning rate and batch size are	The model is trained using labeled data with evaluation metrics such as accuracy, precision, recall, and F1-score. Cross-validation is used to improve robustness.	The model is evaluated based on engagement accuracy and detection consistency rather than traditional classification metrics.	No traditional training is required for rule-based systems. Evaluation is based on relevance and accuracy of recommendations.

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		tuned.			
Model Integration	Tools used	Python, TensorFlow/Keras, OpenCV, MTCNN	Python, TensorFlow/Keras, OpenCV	Python, OpenCV, SORT	Python, MongoDB, Pandas
Model Deployment	Testing Environments	Tested in a local environment using Python and Jupyter Notebook before integration into the system.	Tested locally with validation datasets before deployment and live camera feeds.	Tested using recorded video streams and live camera feeds	Tested using simulated inputs (age, gender, emotion, engagement).
	Deployment Platform	Deployed on a local server or edge device for real-time inference.	Deployed on a local server integrated with the main system.	Deployed on an edge device (kiosk system) for real-time processing.	Deployed within the backend system connected to the database.
	Deployment Method	Exposed as part of a FastAPI-based system for real-time face analysis.	Integrated into the FastAPI backend for real-time emotion prediction.	Runs continuously as part of the real-time inference pipeline.	Provides recommendations through API endpoints in real time.
Future Enhancements	Model Improvement:	Improve accuracy using larger datasets and optimize model for faster inference using lightweight architectures.	Incorporate more diverse datasets and improve real-time performance using optimized architectures.	Improve accuracy using advanced tracking methods and integrate additional behavioral features	Implement machine learning-based recommendation systems and personalize ads using user history

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				such as gesture recognition.	and behavior patterns.
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